

1. Most Effective Imputation Strategy

Among all the imputation techniques, MICE (Multiple Imputation by Chained Equations) was found to be the most effective. It considers relationships between multiple features and generates more accurate and realistic values compared to simple methods like mean/median. While KNN also performed well, MICE provided better consistency across all numerical variables

2. Best Outlier Handling Method

Among the outlier handling techniques, Winsorization was the most effective in preserving data quality. Unlike methods that remove data points, Winsorization caps extreme values within a specified range, ensuring no data loss. It maintained the dataset size while reducing the impact of extreme values, making it more suitable for this dataset.

3. Impact of Data Cleaning

Data cleaning significantly improved the usability of the dataset. Missing values were properly handled, ensuring no incomplete records remained. Outliers were controlled, resulting in a more stable and consistent data distribution. Overall, the dataset became more reliable and suitable for further analysis and machine learning tasks.